

Home automation using BLYNK

Project Aim

Design and implement an ESP32-based IoT room automation system using Blynk Cloud and Wi-Fi.

Project Overview

This project uses an ESP32 connected to the Blynk Cloud to remotely control a room light and fan through a smartphone. An LED represents the room light and a Servo motor represents the fan. Separate Blynk button widgets allow independent control over each device using real-time Wi-Fi communication.

Components Used

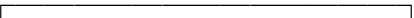
- ESP32 DevKit V1
- LED (Room Light)
- 220 Ω Resistor
- Servo Motor (180 degree)
- Jumper Wires
- Blynk Mobile App
- Wi-Fi Connection

Construction and Wiring Details

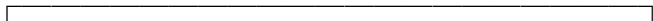
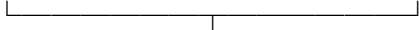
Component	Component Pin	Connected To (ESP32)
LED	Anode (+)	GPIO 2 through 220 Ω Resistor
LED	Cathode (-)	GND
220 Ω Resistor	One End	GPIO 2

Component	Component Pin	Connected To (ESP32)
220Ω Resistor	Other End	LED Anode (+)
Servo Motor	Signal (Orange/Yellow)	GPIO 4
Servo Motor	VCC (Red)	5V
Servo Motor	GND (Brown/Black)	GND

Code Structure / Program Flow

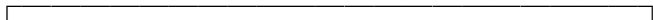


| START |

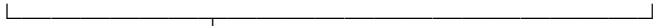


| Initialize ESP32, |

| LED and Servo Pins |



| Connect to Wi-Fi |





|

| Connect to Blynk |

| Cloud |

|

|



|

| Wait for Commands |

| from Blynk App |

|

|



|

| V0 Button Changed? |

|

| Yes | No

▼ |

| |

| Light ON? | |

| |

| Yes | No |

▼ ▼ |

| | |

| LED | | LED | |

| ON | | OFF | |

| |

| | |

| |

|

▼

|

| V1 Button Changed? |

| Yes | No

▼ |

| Fan ON? | |

| Yes | No |

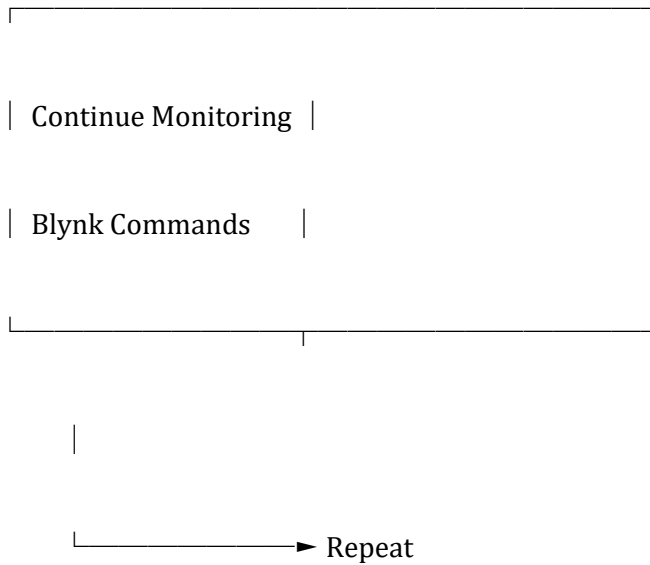
▼ ▼ |

| Servo to | | Servo to |

| 180° | | 0° |

| |

|



Source Code

Diagram.jetson

```
{
  "version": 1,
  "author": "Aman Abdulla",
  "editor": "wokwi",
  "parts": [
    { "type": "board-esp32-devkit-c-v4", "id": "esp", "top": 134.4, "left": -23.96, "attrs": {} },
    { "type": "wokwi-led", "id": "led1", "top": -42, "left": 128.6, "attrs": { "color": "red" } },
    {
      "type": "wokwi-resistor",
      "id": "r1",
      "top": 72,
      "left": 124.25,
      "rotate": 90,
```

```

    "attrs": { "value": "1000" }
  },
  {
    "type": "wokwi-servo",
    "id": "servo1",
    "top": -39.4,
    "left": -247.8,
    "rotate": 270,
    "attrs": {}
  }
],
"connections": [
  [ "esp:TX", "$serialMonitor:RX", "", [] ],
  [ "esp:RX", "$serialMonitor:TX", "", [] ],
  [ "esp:2", "r1:2", "green", [ "h76.8", "v-9.6" ] ],
  [ "r1:1", "led1:A", "green", [ "h0" ] ],
  [ "esp:GND.2", "led1:C", "black", [ "v0", "h67.2" ] ],
  [ "servo1:PWM", "esp:4", "red", [ "v0", "h221", "v172.8" ] ],
  [ "servo1:V+", "esp:5V", "orange", [ "v0" ] ],
  [ "servo1:GND", "esp:GND.2", "black", [ "v28.8", "h259.2", "v28.8" ] ]
],
"dependencies": {}
}

```

Sketch.ino code

#define BLYNK_TEMPLATE_ID "TMPL3SCi5p1Rn"

#define BLYNK_TEMPLATE_NAME "HOME AUTOMATION 1"

#define BLYNK_AUTH_TOKEN "QApJjR95ltvV01JnFZV30bPeXRpxPPq7"

#define BLYNK_PRINT **Serial**

#include <WiFi.h>

#include <WiFiClient.h>

#include <BlynkSimpleEsp32.h>

#include <ESP32Servo.h>

char ssid[] = "Wokwi-GUEST";

char pass[] = "";

#define LIGHT_PIN 2

#define SERVO_PIN 4

Servo fanServo;

BLYNK_WRITE(V0)

{

int lightState = param.asInt();

if (lightState)

{

digitalWrite(LIGHT_PIN, HIGH);

Serial.println("Light ON");

```
    }  
    else  
    {  
        digitalWrite(LIGHT_PIN, LOW);  
        Serial.println("Light OFF");  
    }  
}
```

```
BLYNK_WRITE(V1)  
{  
    int fanState = param.asInt();  
  
    if (fanState)  
    {  
        fanServo.write(180); // Fan ON  
        Serial.println("Fan ON");  
    }  
    else  
    {  
        fanServo.write(0); // Fan OFF  
        Serial.println("Fan OFF");  
    }  
}
```

```
void setup()  
{
```

Serial.begin(115200);

pinMode(LIGHT_PIN, OUTPUT);

digitalWrite(LIGHT_PIN, LOW);

fanServo.attach(SERVO_PIN);

fanServo.write(0);

Blynk.begin(BLYNK_AUTH_TOKEN, ssid, pass);

}

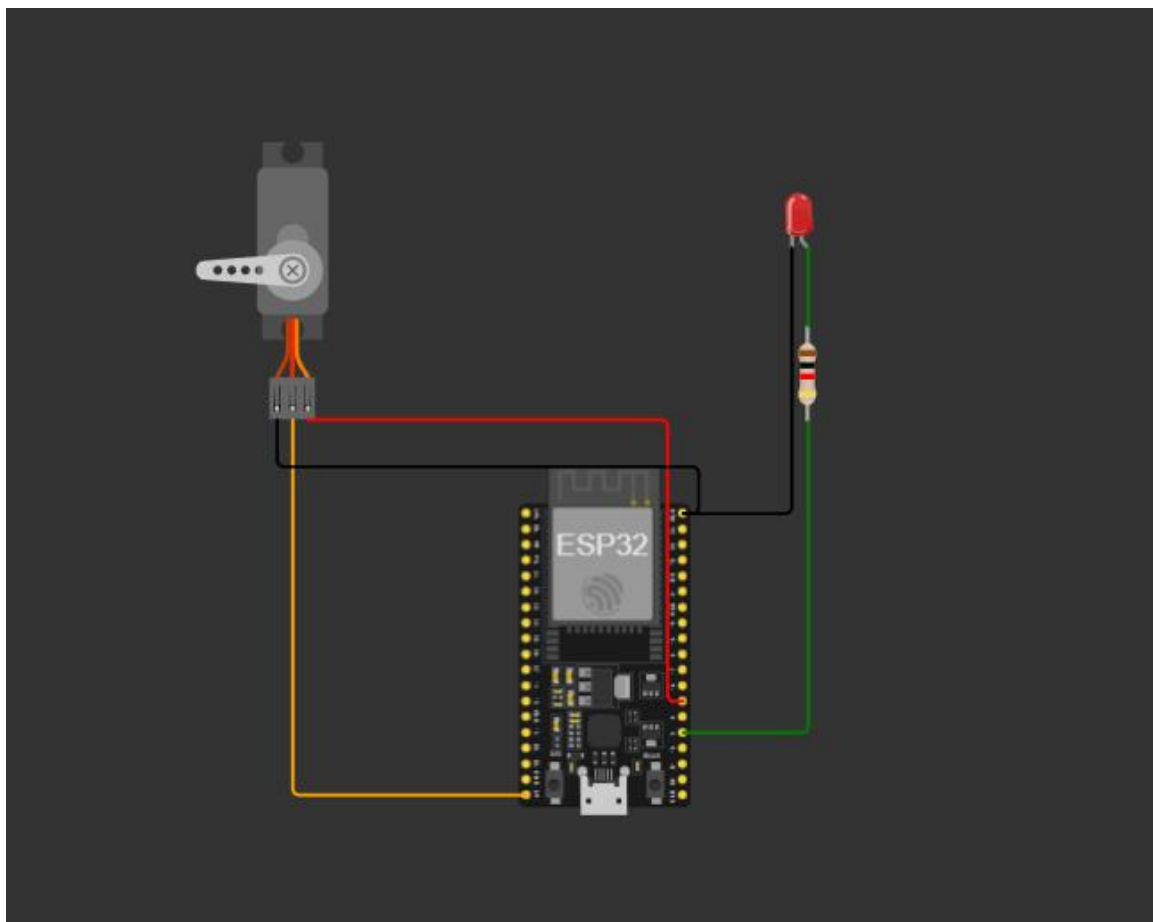
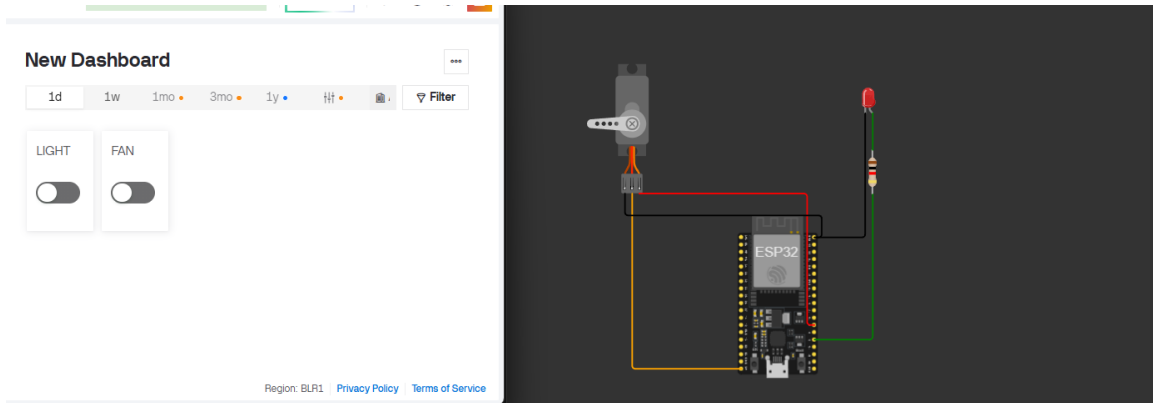
void loop()

{

Blynk.run();

}

Project Photograph



Links

Google drive :

<https://drive.google.com/drive/folders/1Hasoy5jrZQSUM9kxFmAcLoFnFg3UZ5w1?usp=sharing>

Wokwi : <https://wokwi.com/projects/466371165892122625>

Git hub repo : <https://github.com/AmanAZ5950/CIRCUITRON.git>